

Q. 7. What is a synchronous capacitor ?

Ans. An overexcited synchronous motor is called synchronous capacitor, because, like a capacitor, it takes a leading current.

Q. 8. What are the causes of faulty starting of a synchronous motor ?

Ans. It could be due to the following causes :

1. voltage may be too low – at least half voltage is required for starting
2. there may be open-circuit in one phase – due to which motor may heat up
3. static friction may be large – either due to high belt tension or too tight bearings
4. stator windings may be incorrectly connected
5. field excitation may be too strong.

Q. 9. What could be the reasons if a synchronous motor fails to start ?

Ans. It is usually due to the following reasons :

1. voltage may be too low
2. some faulty connection in auxiliary apparatus
3. too much starting load
4. open-circuit in one phase or short-circuit
5. field excitation may be excessive.

Q. 10. A synchronous motor starts as usual but fails to develop its full torque. What could it be due to ?

Ans. 1. exciter voltage may be too low 2. field spool may be reversed 3. there may be either open-circuit or short-circuit in the field.

Q. 11. Will the motor start with the field excited ?

Ans. No.

Q. 12. Under which conditions a synchronous motor will fail to pull into step ?

Ans. 1. no field excitation 2. excessive load 3. excessive load inertia

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1. In a synchronous motor, damper winding is provided in order to
 - (a) stabilize rotor motion
 - (b) suppress rotor oscillations
 - (c) develop necessary starting torque
 - (d) both (b) and (c)
2. In a synchronous motor, the magnitude of stator back e.m.f. E_b depends on
 - (a) speed of the motor
 - (b) load on the motor
 - (c) both the speed and rotor flux
 - (d) d.c. excitation only
3. An electric motor in which both the rotor and stator fields rotates with the same speed is called a/anmotor.
 - (a) d.c.
 - (b) chrage
 - (c) synchronous
 - (d) universal
4. While running, a synchronous motor is compelled to run at synchronous speed because of
 - (a) damper winding in its pole faces
 - (b) magnetic locking between stator and rotor poles
 - (c) induced e.m.f. in rotor field winding by stator flux
 - (d) compulsion due to Lenz's law
5. The direction of rotation of a synchronous motor can be reversed by reversing
 - (a) current to the field winding
 - (b) supply phase sequence
 - (c) polarity of rotor poles
 - (d) none of the above

6. When running under no-load condition and with normal excitation, armature current I_a drawn by a synchronous motor
 - (a) leads the back e.m.f. E_b by a small angle
 - (b) is large
 - (c) lags the applied voltage V by a small angle
 - (d) lags the resultant voltage E_R by 90° .
 7. The angle between the synchronously-rotating stator flux and rotor poles of a synchronous motor is called..... angle.
 - (a) synchronizing
 - (b) torque
 - (c) power factor
 - (d) slip
 8. If load angle of a 4-pole synchronous motor is 8° (elect), its value in mechanical degrees is
 - (a) 4
 - (b) 2
 - (c) 0.5
 - (d) 0.25
 9. The maximum value of torque angle α in a synchronous motor isdegrees electrical.
 - (a) 45
 - (b) 90
 - (c) between 45 and 90
 - (d) below 60
 10. A synchronous motor running with normal excitation adjusts to load increases *essentially* by increase in its
 - (a) power factor
 - (b) torque angle
 - (c) back e.m.f.
 - (d) armature current.
 11. When load on a synchronous motor running with normal excitation is increased, armature current drawn by it increases because
 - (a) back e.m.f. E_b becomes less than applied voltage V
 - (b) power factor is decreased
 - (c) net resultant voltage E_R in armature is increased
 - (d) motor speed is reduced
 12. When load on a normally-excited synchronous motor is increased, its power factor tends to
 - (a) approach unity
 - (b) become increasingly lagging
 - (c) become increasingly leading
 - (d) remain unchanged.
 13. The effect of increasing load on a synchronous motor running with normal excitation is to
 - (a) increase both its I_a and p.f.
 - (b) decrease I_a but increase p.f.
 - (c) increase I_a but decrease p.f.
 - (d) decrease both I_a and p.f.
 14. Ignoring the effects of armature reaction, if excitation of a synchronous motor running with constant load is increased, its torque angle must necessarily
 - (a) decrease
 - (b) increase
 - (c) remain constant
 - (d) become twice the no-load value.
 15. If the field of a synchronous motor is under-excited, the power factor will be
 - (a) lagging
 - (b) leading
 - (c) unity
 - (d) more than unity
 16. Ignoring the effects of armature reaction, if excitation of a synchronous motor running with constant load is decreased from its normal value, it leads to
 - (a) increase in but decrease in E_b
 - (b) increase in E_b but decrease in I_a
 - (c) increase in both I_a and p.f. which is lagging
 - (d) increase in both I_a and ϕ
 17. A synchronous motor connected to infinite bus-bars has at *constant* full-load, 100% excitation and unity p.f. On changing the excitation only, the armature current will have
 - (a) leading p.f. with under-excitation
 - (b) leading p.f. with over-excitation
 - (c) lagging p.f. with over-excitation
 - (d) no change of p.f.
- (Power App.-II, Delhi Univ. Jan 1987)
18. The V-curves of a synchronous motor show relationship between
 - (a) excitation current and back e.m.f.
 - (b) field current and p.f.
 - (c) d.c. field current and a.c. armature current
 - (d) armature current and supply voltage.

19. When load on a synchronous motor is increased, its armature currents is increased provided it is
 (a) normally-excited
 (b) over-excited
 (c) under-excited
 (d) all of the above
20. If main field current of a salient-pole synchronous motor fed from an infinite bus and running at no-load is reduced to zero, it would
 (a) come to a stop
 (b) continue running at synchronous speed
 (c) run at sub-synchronous speed
 (d) run at super-synchronous speed
21. In a synchronous machine when the rotor speed becomes more than the synchronous speed during hunting, the damping bars develop
 (a) synchronous motor torque
 (b) d.c. motor torque
 (c) induction motor torque
 (d) induction generator torque
22. In a synchronous motor, the rotor Cu losses are met by
 (a) motor input
 (b) armature input
 (c) supply lines
 (d) d.c. source
23. A synchronous machine is called a doubly-excited machine because
 (a) it can be overexcited
 (b) it has two sets of rotor poles
 (c) both its rotor and stator are excited
 (d) it needs twice the normal exciting current.
24. Synchronous capacitor is
 (a) an ordinary static capacitor bank
 (b) an over-excited synchronous motor driving mechanical load
 (c) an over-excited synchronous motor running without mechanical load
 (d) none of the above

(Power App.-II, Delhi Univ. Jan. 1987)

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ANSWERS

1. d 2. d 3. c 4. b 5. b 6. c 7. b 8. a 9. b 10. d 11. c
 12. b 13. c 14. a 15. a 16. d 17. b 18. c 19. d 20. b 21. d 22. d
 23. c 24. c